

Evaluating Depth Perception in Augmented Reality X-Ray Vision

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ABSTRACT

Depth perception in AR describes the ability to perceive depth from AR-generated objects. This functionality enables an emergent capability: x-ray vision, or visualizing objects past an occluding surface. To evaluate x-ray vision's feasibility, we propose experiments that depict a virtual object beyond a solid wall. In one condition this display is mitigated with a semi-transparent virtual window, while in another condition no such effect is presented. Depth estimates will be measured using triangulation by walking. Our motivation is to understand discrepancies between perceived distances of virtual and real-world objects so that AR systems can accurately display objects in the real-world.

Index Terms: Augmented Reality—Depth Perception—Optical see-through Display—X-Ray Vision

1 INTRODUCTION

X-ray vision can be used to gather visual information about the environment past an opaque surface. An ongoing concern with x-ray vision is the lack of an equivalent perceptual trait in the human visual system. When perceiving objects, the human visual system utilizes a collection of visual depth cues. Perceived depth is dependent upon factors such as the object's distance, size, and shape [1]. This collection of visual depth cues, developed in natural environments and situations, does not include an innate capability to perceive depth information beyond opaque surfaces. In fact, occlusion, one of the strongest depth cues [1], provides depth information counter to the actual depth of the visualized object. This raises the question of whether humans can accurately comprehend the depth of objects presented beyond an occluding surface as depicted by AR x-ray vision.

This work is motivated in part by the many circumstances in which AR x-ray vision could be useful in different industries. A primary purpose could be to improve task accuracy, as in the context of surgery. AR x-ray vision could aid doctors in medical imaging tasks required during surgery by supporting a visualization modality that doesn't require translating data from a computer screen to a real-world position [2]. Such an operation would be performed at a near-field distance (2m or less), but the benefits of AR x-ray vision can also apply to medium-field (2m-30m) or far-field (30m or greater) distances [1]. For these tasks, often the advantage of x-ray vision is improved awareness, higher task efficiency, and a more complete understanding of the environment.

Some of these tasks include localizing pipe placements [4], drafting and design, disaster response and recovery, or remote robot navigation, along with many other comparable applications. Such work can involve inherently medium-field actions in which operators navigate a room-sized space in order to accomplish their goals. The potential utility of augmenting these applications, in conjunction with the dearth of medium-field x-ray vision research, motivates a closer look at AR x-ray vision at these distances.

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As such, in this paper, we propose a set of experiments to measure how virtual objects at medium-field distances are perceived when presented with AR x-ray vision in an indoor location.

2 LITERATURE REVIEW

In investigating AR x-ray vision, it is important to discuss our proposed work in the context of prior research. Past work has often shown that depth perception in virtual and augmented reality is misestimated [3, 4, 6]. In contrast, depth perception in full-cue real world contexts is generally veridical [1]. Though there are far more applicable studies than can be cited here, a brief discussion of some may be instructive.

In particular, Phillips [5] describes a set of experiments also investigating the efficacy of medium-field x-ray vision techniques and RW vision through a window, across several different experimental conditions. Findings include no significant differences in observed depth accuracy between judgments to a RW target made through a window and judgments to a virtual target made through an opaque wall. Based on the history of depth misestimations in AR, this is a surprising finding that warrants additional investigation. In particular, do these findings hold in other indoor environments or under differing experimental conditions? Our proposed experiment targets a few potential confounds by using a different head-mounted display, environment, and experimental setup. As a result, some weaknesses in the research could be addressed, and further verification of these results could be established.

In contrast to the above, Gagnon et al. [3] found general depth underestimation in depth judgments to targets between 10 and 35 m in AR (Figure 1). This work describes two experiments that tested the egocentric depth perception of objects in AR. Experiment 1 tested for judgments from near to medium-field distances (up to 35 m). Experiment 2 used verbal reports to gain an absolute measure on perceived objects in AR compared to RW. When judging distances between 10 to 35 m, observers tended to underestimate the distance of virtual objects when compared to RW objects in an indoor setting.

Swan et al. [6] also found general depth misestimations in AR. This work describes two experiments that explore the tendency for underestimation from medium to far-field distances (5 to 45 m). Experiment 1 used a perceptual matching protocol at these distances in the x-ray condition and compared the depth properties of the upper versus lower visual field. The results of this experiment indicated a switch from underestimation to overestimation at ~23 m for AR-generated graphics. Experiment 2 used blind walking to compare differences in AR depth judgement between RW targets, virtual targets, and a combination of both kinds of objects in a single environment. These objects were all judged at distances between 3 to 7 meters.

Given this background and foundation, it seems clear that several prior experiments with depth perception in AR have found general depth misestimation across a variety of contexts (Figure 1) [3, 4, 6]. Considering the contrasting research [5], it is important to test this observation, especially in the unique context of x-ray vision.

3 METHODS

In our first experimental condition, after an informed consent session, participants will wear an augmented reality optical see-through display, such as the HoloLens 2 or the Magic Leap 2. They will

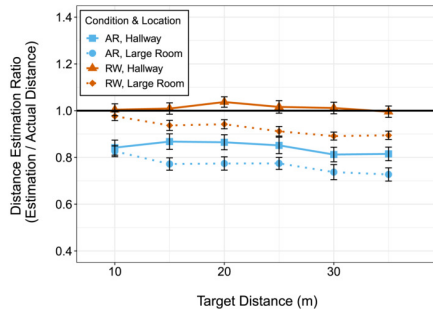


Figure 1: A graph displaying measured distance estimation ratio (estimation / actual distance) given target distance in Gagnon et al. The horizontal black line indicates perfect accuracy. The left graph shows the difference in estimation between the AR condition and RW condition based on the order in which the conditions were viewed. Target distances were consistently underestimated in AR (blue lines) compared to the RW (orange lines), and an effect of the environment was also observed [3].

then go through a series of trials in which the virtual object will be displayed at a medium-field distance. The participant will see this object through a virtual semi-transparent window over a movable opaque wall and will provide a depth estimate to this object through triangulation by walking (Figure 2), a visually-directed action. The virtual target object will be a green cylinder of an equivalent size to a tall soda can. The environment itself will be an indoor setting in a room with enough space for medium-field depth judgments.

In the second experimental condition, participants will perform the same experimental procedures, with the exception that depth estimates will be made to the virtual object through a solid wall with no window. This condition tests the depth accuracy that could be expected from full x-ray vision, with more limited visual cues.

Finally, for the control condition (Figure 3), environmental factors will be kept the same as much as possible. Participants will still wear an AR optical see-through headset; however, these displays will not be active, and instead participants will see a RW object instead of the virtual object. Other components of the test will remain the same, including room type/size, depth measurements collected, and other experimental procedures.

A significant concern cited in previous research was participant memorization of the three primary distances [5]. In order to remove a potential confound, six different distances will be tested. This should be enough to insure that participants will not be able to easily memorize them. Further, a large enough indoor environment will be used such that none of the distances correspond to the edges of the environment, which has been shown to bias depth judgments in prior research [7]. Responses to the perceived distances of the virtual object will also be measured based on a visually-directed action procedure in order to maximize ecological validity; previous research has found that, generally, human performance in visually directed action tasks (such as direct blind walking) is superior to performance for other tasks (such as verbal report), and participant action is the logical corollary of the medium-field tasks motivating this work [1].

The presented distances will be randomized such that no distance will be repeated twice in a row. After each trial in all conditions, participants will be instructed to keep their eyes closed and led obliquely back to the starting location. This is to disrupt any potential feedback the participant might receive from seeing where they walked to or being led directly back to the starting location. The order of the conditions will be counter-balanced using a 3x3 Latin Square as well; this can counteract potential ordering effects, such as the ones seen in Gagnon et al. [3].

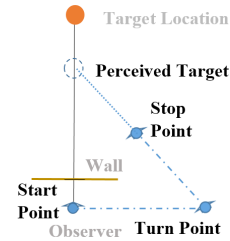


Figure 2: The proposed visually directed action, triangulation by walking. An observer views the target from behind a wall, then closes their eyes and walks obliquely until instructed to stop and turn toward the target. They then walk forward until instructed to stop again. This procedure allows a participant to provide a visually directed distance estimate without the potential confound of walking through a space where they last saw a wall [5].

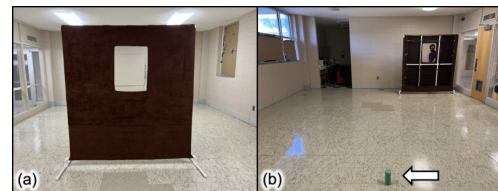


Figure 3: Control condition for our proposed experiment. The observer views the target object behind the opening in the wall. The wall and target object positions are adjustable to allow estimations at various distances [5].

4 CONCLUSION

Given this setup and methodology, the effectiveness of x-ray vision depth perception as compared to depth perception in the real world can be better understood. In the context of ongoing research, these experimental results could support or challenge previous experimental findings, especially by investigating potential confounds. This will help clarify the efficacy AR x-ray vision can have for tasks requiring fine-tuned depth judgments. This knowledge could both advance the scientific understanding of perception and augmented reality and allow novel applications to be developed and implemented.

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